Walls of Memory: Singapore 1942

Walls of Memory: Singapore 1942 *(student proposal)*

> **Assessment:** ◆ Fit with adjustments · **Category:** Immersive Experience · **Assessed:** 2026-07-14 · **Status:** awaiting instructor approval

Proposal (as submitted): A projected experience on the wall, with an app that controls the projected images and allows manipulating the visual effects, to show the impact of the Japanese invasion of Singapore.

Category note: an atmosphere-first projected installation with a companion controller — squarely **Immersive Experience** (nearest instructor example: the Interactive museum exhibit). Runner-up: Education + Training — but the product's core promise is an experience to be felt, not lessons or drills.

Adjustments required for approval:

- The projection renderer must be the team's **own real-time C++ engine** (SDL2/OpenGL scene compositing, layers, particles, shaders at 60 fps) — not a video/slideshow player. "Manipulating the visual effects" must mean **live parameter control of engine effects** (emitter rates, intensity, transitions), never switching pre-rendered clips.
- The **phone ↔ renderer control link is core tech**: a custom local-network protocol (UDP/ENet on the installation LAN) with a measured control-to-photon latency budget (**under 100 ms**). Firebase stays BaaS for content and stats — it must never carry the live control path.
- **Mobile-output policy (Appendix F):** the controller app on the reference phone (Pixel 8a/9a) ships as a **first-class output** — the control deck plus a live on-device companion rendering of the chapters; the projection PC ships only as the installation "(ops)" side of the role split.
- **Historical care:** all exhibit content is data-driven and authored through the curator editor from documented sources; archival imagery/audio licensing

verified; respectful memorial tone with a sensitive-content notice at entry (consider M2 G16 Accessibility & Comfort as an additional pick).

References:

- teamLab — Story of the Forest, National Museum of Singapore (official)
- Changi Chapel and Museum — the Japanese Occupation story (official)
- HeavyM — projection mapping software with live effect control (official)

Core Tech Requirements

Legend: ✔ Applies directly | ◆ Applies with domain adaptation | ● Optional | ✘ Not needed | » Second trimester

Core Tech Requirement	Walls of Memory: Singapore 1942
External Database (Firebase)	✔
Authentication »	●
Analytics »	✔
C++ Performant Service	✔
Real-time C++ Simulation Loop	✔
Application Communication	✔
Continuous Integration	✔
Data-Driven Architecture	✔
Front-end + Local Backend	✔
Custom Tools Development	✔
2D Graphical Output	✔
3D (strong students only)	✘
Networking System	✔
Spatial / Reactive Audio Engine — positional sound, input- and state-driven mixing (new)	✔

How each requirement applies:

> *Platform fit:* **Native mobile (reference device: Google Pixel 8a / 9a) + Native PC (ops)** — the phone is the docent/visitor control deck AND renders a live companion view of the chapters; the PC (ops) side drives the projector wall in the installation.

Front end: Projected scene rendering (layered archival imagery, particles, shader effects, FreeType captions), phone control deck + companion view.
Languages: C++, GLSL.

Back end: Scene & sequence engine (invasion chapters), effect parameter system, control-protocol server, spatial audio, Firebase content/stats.
Languages: C++.
Backend scope: Local only — the renderer + controller pair lives on the installation LAN; Firebase remains BaaS.

- *Simulation loop:* per-frame scene compositing, particle simulation and transition playback, all steered live by control events — the projection is computed, not played back.
- *Data-driven:* chapters, scenes, layers, emitter presets, captions and hotspots defined as JSON — the entire exhibit is content the curator editor can author.
- *Custom tool:* curator chapter/scene editor — historians assemble sequences, place archival imagery and tune effect presets without code.
- *External DB:* exhibit content and versions. *Auth »:* operator/docent login on the control deck. *Analytics »:* per-chapter interaction and dwell statistics.
- *2D:* layered parallax scenes over archival photographs with custom shaders (fire glow, smoke, searchlights). *3D:* not needed. *Networking:* ✔ — the app↔renderer control protocol is core to the concept.
- *Spatial/reactive audio:* sirens crossing the room, bombardment rumble and radio broadcasts positioned per scene, with the mix reacting to chapter state and live control input.

Suggested Tech Goals

G1–G3 ★ are the common goals (fixed for all categories); the other 3 are picked from the Immersive Experience pool in the CS Goals Pool (`milestone-goals-pool-per-category.md`).

M1:

- **G1 ★ Application Shell, Real-time C++ Simulation Loop, Input & Core Logic**
  Tech & dependencies
  - **Tech:** SDL2 fullscreen projection shell with fixed-timestep loop; input events dispatched from touch (dev), keyboard (dev) and the LAN control channel into one event queue; C++17/20, CMake, Git.
  - **Prerequisites:** None — foundation of this project.
  - **Feeds into:** Every goal; M1 G10's sequences and M1 G5's emitters tick inside this loop; the control link (adjustment 2) lands in its input layer until a dedicated pool goal exists.
- **G2 ★ Data-Driven Architecture of Core Objects / Entity System**
  Tech & dependencies
  - **Tech:** Entity/component model for chapters, scenes, layers, emitter presets and hotspots in JSON (nlohmann/json) with hot-reload so curators see edits instantly.
  - **Prerequisites:** G1 (loop & update hooks).
  - **Feeds into:** M1 G4/G5/G10 consume this model; M2 G2 (curator editor) edits it directly.
- **G3 ★ Debugging, Profiling, Stress Test & CI**
  Tech & dependencies
  - **Tech:** ImGui overlays (frame time, particle counts, control-link latency), Tracy profiling, Catch2 tests, long soak runs for all-day unattended installation running, GitHub Actions Windows+Linux CI.
  - **Prerequisites:** G1 (loop to instrument).
  - **Feeds into:** M2 G3 (packaging CI); the latency budget in adjustment 2 is measured here.
- **G4 2D Scene Rendering & Animation System** — layered archival-imagery scenes with tweened camera and light moves: the wall itself.

Tech & dependencies

- **Tech:** OpenGL renderer with GLSL, parallax layers over archival photographs, tween animation, FreeType captions sized for projection distance, stb_image.
- **Prerequisites:** G1 (loop), G2 (scene data).
- **Feeds into:** M1 G5 (particles composite through it), M1 G10 (sequences animate it), M2 G8 (advanced effects extend it).

- **G5 Particle & VFX System** — the manipulable effects palette: fire, smoke, falling leaflets, searchlights, rain.
  Tech & dependencies
  - **Tech:** OpenGL-instanced particles with emitter presets as JSON data; every parameter (rate, intensity, direction, color) adjustable at runtime — the surface the app manipulates.
  - **Prerequisites:** M1 G4 (renderer).
  - **Feeds into:** M2 G8 (showcase effects), the live-control demo at the wk9 checkpoint.
- **G10 Narrative / Sequence Engine** — the invasion timeline as data: air raids → the fall → occupation, chapter by chapter.
  Tech & dependencies
  - **Tech:** Data-defined chapter sequences (scene order, timings, transitions, audio cues) with play/pause/scrub/jump; deterministic playback for reliable exhibition runs.
  - **Prerequisites:** G1 (tick), G2 (chapter data).
  - **Feeds into:** M2 G2 (the editor authors sequences), M2 G7 (audio cues ride the sequence), the memorial arc Design owns.

Proposed new pool goal (requires instructor approval): *Remote Control & Companion Link* — a networked control protocol (discovery, session, low-latency parameter/command channel) between a companion app and the experience renderer. Until approved, the minimum control channel is delivered inside G1 ★'s input layer.

M2:

- **G1 ★ Architecture — Optimization & Platform Validation**
  Tech & dependencies
  - **Tech:** Tracy-guided optimization of compositing and particles; ASan on MSVC; companion build validated on the reference phone (Pixel 8a/

9a), projection build validated at 1080p60 with the full effects palette; control-to-photon latency measured under 100 ms.
  - **Prerequisites:** All M1 goals.
  - **Feeds into:** The showcase installation + companion build every other M2 goal demos on.
- **G2 ★ Content Editor — Core** — the curator editor: chapters, scenes, layers and effect presets authored without code.
  Tech & dependencies
  - **Tech:** Desktop authoring tool editing chapters/scenes/layers/presets with JSON round-trip, undo/redo and live preview through the project's renderer.
  - **Libraries/Software:** Dear ImGui, nlohmann/json, project's SDL2/OpenGL renderer (live preview)
  - **Prerequisites:** M1 G2 (data model), M1 G10 (sequence schema).
  - **Feeds into:** Every authored chapter; adjustment 4's "historians author without code" requirement.
- **G3 ★ CI — CMake, Code Quality, Build Standards & Packaging**
  Tech & dependencies
  - **Tech:** CMake presets for renderer, companion app and editor; clang-format + clang-tidy gates; one-command packaging of the ops build, the mobile build and the editor; CI smoke test plays a full chapter headless.
  - **Prerequisites:** M1 G3.
  - **Feeds into:** Showcase/installation builds; deterministic-playback guard for G10.
- **G7 Spatial / Reactive Audio Engine** — the purple requirement: the soundscape of the fall of Singapore, positioned and reactive.
  Tech & dependencies
  - **Tech:** Positional mixing (FMOD/OpenAL) — sirens traversing the field, bombardment rumble, radio voices per scene; mix reacts to chapter state and live control input.
  - **Prerequisites:** M1 G10 (sequence cues), CS audio foundation.
  - **Feeds into:** TEAM TA1 ★ (M2) grading; the emotional weight of the experience.
- **G8 Particle / VFX II** — the showcase effects: era-accurate impact visuals and chapter transitions, manipulated live from the app.

Tech & dependencies

  - **Tech:** Advanced compositing (burn/ash/fade transitions, searchlight volumetrics, rain over rooftops), every showcase effect exposing live parameters to the control deck within the latency budget.
  - **Prerequisites:** M1 G5 (particle base), M2 G1 ★ (frame budget).
  - **Feeds into:** The core demo moment — a docent reshaping the wall in real time.

- **G13 Mobile Platform Port** — the companion app as a first-class output: control deck + on-device chapter rendering.
  Tech & dependencies
  - **Tech:** Android (NDK) build of the engine's companion view rendering all chapters on-device; touch control deck driving every live effect parameter; interruption-safe control sessions.
  - **Prerequisites:** M2 G1 ★ (validated build), M1 G5/G10 (what it controls).
  - **Feeds into:** Adjustment 3 compliance — the mobile target as a real output, not a remote.

(Strong alternate pick: G16 Accessibility & Comfort Options — sensitive-content notice, flash/intensity limits, captions.)

Track Goals & Team Composition

Recommended composition: 6 CS + 1 Design + 2 Art (9) — the museum-exhibit precedent applies: historical scene art, animation and VFX rival the Games surface; a solo designer owns the experience arc and the control-deck UX.

> **Track goals** — suggested Design/Art picks and TEAM notes for this project; deliverables & quality ladders live in the Design, Art and TEAM pool documents.
>
> - **Design M1:**
>   - D1 ★ Experience DD — the memorial arc across invasion chapters, projection space + visitor flow, content-sensitivity plan, feasibility vs. the CS engine plan
>   - D2 Prototype — one chapter end to end: phone controls → wall reacts (greybox/Figma acceptable)
>   - TU1 ★ UI Standards & Wireframe Baseline — both surfaces: wall (attract state, caption readability at distance) and control-deck flows
> - **Design M2:**
>   - D1 ★ UX validation with visitors; final polished control-deck UI + wall entry/attract states on the mobile target
>   - D9 Narrative Presentation — chapters delivered with motion (parallax, reveals, light) — never static text walls
>   - TU1 ★ Functional UI & Usability — Immersive screens, diegetic-first on the wall (entry/attract · pause/exit · settings incl. volume and sensitive-content notice · instructions) + a first-time docent operates the control deck unaided
> - **Art M1:**
>   - 2 artists recommended (historical environment + VFX surface) — A1 ★ Art bible (archival-imagery treatment, era palette; mockups: wall chapter + control deck)
>   - A3 Environment, Props & VFX concepts — 1942 streetscapes, landmarks, war machines
>   - A5 Animation Foundations — scene motion tests (smoke drift, flag, fire cycles)

    - A6 Early Asset Production — first chapter's assets in the prototype by wk8
- **Art M2:**
    - A1 ★ Critical asset set implemented in the showcase build
    - A3 Environment alive — living scenes: rain, smoke, flickering light
    - A5 VFX — bombardment, fire, searchlights: the manipulable palette
    - A6 Cutscenes/Storyboards — chapter transitions storyboarded and rendered
- **TEAM M1:**
    - TA1 ★ Audio Direction & Plan — wartime soundscape direction (sirens, radio, bombardment, silence of aftermath), archival-audio sourcing/licensing, spatial plan against the CS audio system
- **TEAM M2:**
    - TA1 ★ Audio Integration & Quality — Immersive floor: full Games floor (≥1 BGM + ≥6 SFX types) + the Spatial/Reactive Audio purple requirement; reactive moments graded from the gameplay recording
    - TP1 ★ Final Presentation — 7 min in front of class + instructors: live custom-tools showcase, prototype demo, team post-mortem (wk14)

Generated by the Project Proposal Assessor skill · grounding: live folder · references verified 2026-07-14.